NeuroSight AI — Longitudinal MRI Dementia Analysis

Final project report · OASIS-2 · binary dementia-status classification

Final system NeuroSight-E5 · model card 2026-09-17 · report generated 2026-10-21

Research use only — not a medical device. Not a diagnosis. Not for clinical decision-making.

Every number in this report is computed from committed run artifacts (reports/excellence/bakeoff_e5.json, reports/csv/cv_*_mri_only.json, reports/csv/excellence/excellence_S*.json, reports/csv/search_*.json, reports/data_audit.json, MODEL_CARD.md). The generator is scripts/build_reports.py.

1. Executive summary

NeuroSight AI is a research system for binary dementia-status classification (non-demented vs demented) from longitudinal brain MRI on OASIS-2. The final system, NeuroSight-E5, averages five seed-trained instances of the champion architecture (7_dual_attn: per-slice CNN with spatial and channel attention, BiLSTM over visits, additive temporal attention) and applies temperature scaling fitted on the selection set. It was measured once, on the held-out test set, in a single pre-declared bake-off.

Metric	NeuroSight-E5 (held-out test)
Accuracy	59.1% [95% CI 36.4, 77.3]
AUC-ROC	0.777
F1 (demented)	0.571
Sensitivity / specificity	0.545 / 0.636
ECE (temperature-scaled)	0.227
Fitted temperature	0.657
Held-out test visits	22

The held-out test set contains 22 visits, so every test statistic above is noisy: the 95% confidence interval for accuracy covers a 40.9-point range. Five-fold subject-level cross-validation (champion mean AUC 0.744 ± 0.096) is the primary evidence; the table above is the single declared test measurement. Cross-checks that failed during the project are reported, not removed: see sections 4 to 6.

2. Dataset and methodology

OASIS-2 (Washington University Knight ADRC) provides longitudinal T1-weighted MRI for 150 subjects aged 60-96, with 373 imaging sessions. The committed audit matches 373/373 CSV rows to scans on disk, with no missing or empty sessions. Session counts by research group: 190 nondemented, 146 demented, 37 converted (subject counts: 72 nondemented, 64 demented, 14 converted).

Property	Value
Subjects	150 (aged 60-96)
MRI sessions	373 (audit matched 373/373)
Subject groups	72 / 64 / 14 (nondemented / demented / converted)
Labels	Binary per visit from CDR (0 vs above 0), group verified
Splits	Subject-level 70/15/15 + 5-fold subject-level CV (seed 42)

Splits are subject-level: 70/15/15 train/validation/test with 5-fold subject-level cross-validation on the training portion (seed 42). Leakage guards run in the automated test suite: zero subject overlap across splits, and converted subjects never straddle a split, so no patient contributes to two sides of any evaluation. Labels are derived per visit from the clinical dementia rating (CDR 0 vs above 0) and verified against the research group field; converted subjects are split into pre- and post-conversion

sequences.

The headline input mode is mri_only: all clinical metadata is masked out. MMSE and CDR define the label, so feeding them as features would be leakage; the mri_demo and mri_clinical modes exist for comparison, change comparability, and are flagged in the interface. Preprocessing: z-score normalisation, the 32 middle axial slices at 128x128 pixels, from Analyze 7.5 source volumes.

3. Cross-validation ablation study (11 variants)

Five-fold subject-level cross-validation for all eleven ablation variants (mri_only), sorted by mean AUC. The champion variant 7_dual_attn carries forward into the excellence phase. Fold-to-fold standard deviations of 0.075 to 0.180 dwarf most between-variant gaps, so the ranking is indicative rather than definitive.

Model	Mean CV AUC	Std	Per-fold val AUC	Folds
7. DualAttn (champion)	0.744	0.096	0.772/0.636/0.868/0.817/0.629	5
10. 3D-CNN+LSTM	0.732	0.098	0.763/0.785/0.824/0.746/0.543	5
8. Transformer	0.727	0.075	0.768/0.708/0.819/0.746/0.595	5
5. +TemporalAttn	0.725	0.141	0.665/0.846/0.863/0.772/0.481	5
11. ResNet+LSTM	0.710	0.172	0.634/0.821/0.890/0.795/0.410	5
1. CNN-avg	0.689	0.120	0.636/0.600/0.890/0.759/0.562	5
6. +SE	0.689	0.151	0.786/0.595/0.885/0.728/0.452	5
9. 3D-CNN	0.688	0.077	0.719/0.636/0.797/0.719/0.571	5
4. +SpatialAttn	0.688	0.134	0.790/0.662/0.830/0.710/0.448	5
3. CNN+LSTM	0.685	0.118	0.737/0.590/0.846/0.737/0.514	5
2. CNN-last	0.665	0.180	0.848/0.538/0.830/0.728/0.381	5

All eleven variants were trained before the training-bug fixes described in section 6 (v1 lineage); numbers are retained as baselines.

4. Excellence phase: system comparison and negative results

The excellence phase compared five evaluation systems on the same five folds. Selection used mean CV AUC only (tie broken by calibration error); the held-out test set was not touched during this phase. S0 is the champion baseline; S1a scores each fold only with its own fold checkpoint; S1b repeats that for the transformer variant; S2 averages the two variants' predictions within each fold; S3 adds test-time augmentation.

System	Description	Mean AUC	Std	ECE raw	ECE scaled	AUC vs S
S0	Champion v1, single model	0.7444	0.0965	N/A*	N/A*	0.0000
S1a	Same-fold CV, champion checkpoints	0.7444	0.0965	0.1998	0.2512	0.0000
S1b	Same-fold CV, transformer checkpoints	0.7270	0.0750	0.3209	0.1784	-0.0174

System	Description	Mean AUC	Std	ECE raw	ECE scaled	AUC vs S
S2	Cross-variant ensemble (champion + transformer)	0.7381	0.0819	0.2445	0.3069	-0.0063
S3	S2 plus test-time augmentation	0.7311	0.0873	0.2486	0.2797	-0.0132

S0 equals S1a by construction (same checkpoints, same scoring). * S0 ECE was not measured in the initial 5-fold CV ablation artifact; measured as 0.1998 raw / 0.2512 scaled under S1a. Selection rule: max mean CV AUC, tie by ECE, test set untouched (plan guardrails 1 and 2).

4.1 Negative results (attempted, measured, dropped)

Three attempted improvements were dropped under the pre-registered selection rule. Cross-variant ensembling (S2) lost 0.006 AUC against S0 (0.738 vs 0.744): the two variants' errors are too correlated for their average to help. Test-time augmentation (S3) lost 0.013 (0.731 vs 0.744): medically plausible transforms added no information the network had not already used. Hyperparameter search over 8 configurations (learning rate, dropout, LSTM hidden size; 3-fold proxy CV) returned as its top configuration learning rate 0.0003, dropout 0.5, hidden 32 (mean AUC 0.714 ± 0.124) — exactly the deployed configuration, so no change was made.

4.2 Five-seed variance

E5 members are the final models trained with seeds 42 to 46 (s42, s43, s44, s45, s46). The ensemble's test predictions match the seed-42 single model exactly (McNemar chi-squared 0.0, p = 1.0), and the bootstrap mean accuracy (59.2%) sits on the point estimate (59.1%): at this data scale, seed-to-seed variation does not move the test result. The dominant variance term is the fold-to-fold spread (champion CV std 0.096), which is the reason cross-validation, not the single test touch, carries the conclusions.

5. Final bake-off: NeuroSight-E5 vs champion v1

Plan guardrail 1 allowed exactly one further touch of the held-out test set: the declared final bake-off. The declared system was NeuroSight-E5 (5-seed ensemble), the five-seed ensemble of the champion variant.

Metric	NeuroSight-E5	Champion v1 (seed 42) *
Accuracy	59.1% [95% CI 36.4, 77.3]	59.1%
AUC-ROC	0.777	0.785
F1 (demented)	0.571	0.571
Sensitivity	0.545	0.545
Specificity	0.636	0.636
ECE (temperature-scaled)	0.227	0.304
Fitted temperature	0.657	0.375

* v1-lineage checkpoints predate the training-bug fixes in section 6; retained as baselines. McNemar's test on the 22 paired predictions: chi-squared 0.0, p = 1.0 — identical decisions on every test case.

5.1 Calibration

Temperature scaling fitted on the selection set (T = 0.657) reduced test ECE from the v1 lineage's 0.304 (0.348 in the earlier champion evaluation) to 0.227; the raw ensemble without scaling sits at 0.273. Both remain above the pre-registered 0.10 target, so probabilities should be read as rankings, not calibrated percentages.

5.2 Conformal prediction: an honest failure

Split-conformal prediction sets were implemented with a 90% coverage target and measured 54.5% coverage (mean set size 0.864) at the available calibration size; empty sets occur. Conformal output was therefore not deployed and is recorded as a failure in the model card and in reports/excellence/.

6. Methodological disclosures

Every number in this report was produced after the events below; readers should weigh the results accordingly. Items 1 to 4 are the training-bug lineage from MODEL_CARD.md section 5; all are fixed in code (see git history).

- **Cache bug skipped augmentation (fixed):** the v1 ablation trained with a preprocessing-cache bug that silently skipped data augmentation; all v1-lineage baselines in this report inherit it.
- **Broken 3D gradient accumulation (fixed):** the 3D variants trained with broken gradient accumulation, so their effective batch size differed from the configured one.
- **ResNet batch-norm drift (fixed):** the frozen ResNet backbone's batch-norm statistics drifted during fine-tuning.
- **Partly in-sample final-model validation (fixed):** final-model validation was partly in-sample before the fix; E5 members s43-s46 were trained after all fixes above, while v1-lineage numbers are retained as baselines.

- **Leaky leave-one-fold-out ensemble caught by guardrail:** the first excellence implementation ensembled checkpoints across folds, which scores training data. Its first real run produced S1 AUC 0.956, above the honest band for subject-wise-split OASIS-2 studies (66-90% accuracy), and the guardrail caught it before any number was recorded; the design was replaced with same-fold ensembling (commit a4d75f2).

7. Interpretability (Grad-CAM)

Gradient-weighted class activation mapping highlights the slice regions that drive each prediction. The committed figure below is generated by the evaluation pipeline for the champion architecture on a held-out test visit; the generator embeds it only if the file exists and never substitutes a synthetic image.

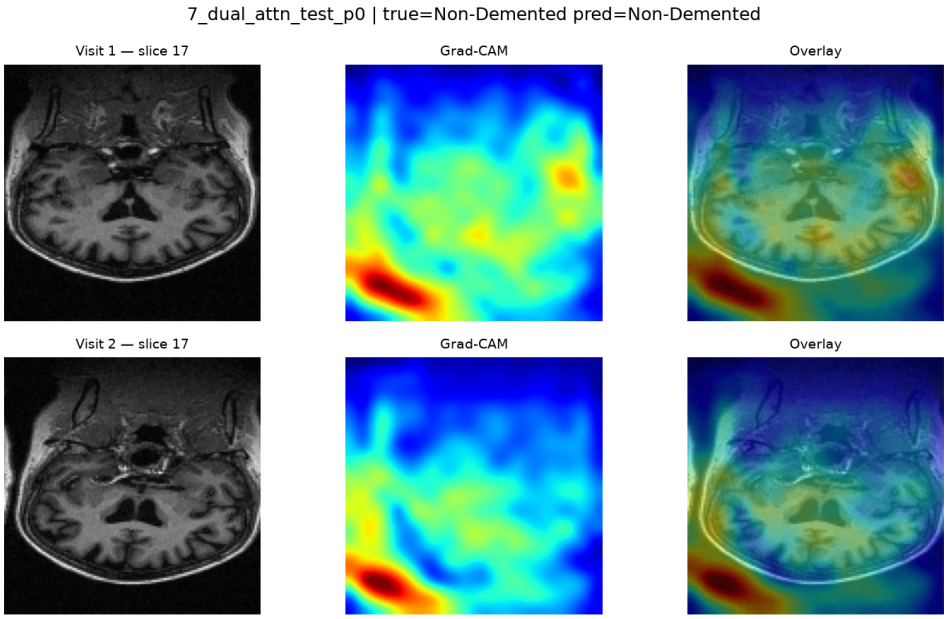


Figure: Grad-CAM overlay for the champion architecture (7_dual_attn) on a held-out test visit, from reports/figures/gradcam/. Warm regions are the evidence the model used.

8. Threats to validity, limitations, future work

- **Small held-out set:** 22 visits; the accuracy CI covers a 40.9-point range, so test numbers describe this split, not a population.
- **Single cohort and scanner family:** OASIS-2 only, 1.5T Analyze-format scans; no external validation yet.
- **Missed cases:** sensitivity 0.545 at the default threshold means 45.5% of demented visits are missed; the system is not a screening instrument.
- **Uncertainty outputs are indicative only:** ECE 0.227 is above the 0.10 target and conformal coverage (54.5%) is below the 90% target.
- **Lineage mixing:** v1-lineage baselines predate the bug fixes, so comparisons across the fix boundary are indicative.
- **Retraining variance:** fold-to-fold std of ±0.096 AUC means a retrain can move numbers by points; labels are binary only (no CN/MCI/AD staging).

Future work follows docs/plan/07-MODEL-EXCELLENCE-PLAN.md: pre-training on OASIS-3 before fine-tuning on OASIS-2, external validation on ADNI/AIBL, conformal sets revisited at a calibration size that can support the coverage target, calibration beyond a single temperature, and multi-class staging.

References

[1] Marcus, D. S., Fotenos, A. F., Csernansky, J. G., Morris, J. C., & Buckner, R. L. (2010). Open Access Series of Imaging Studies (OASIS): longitudinal MRI data in nondemented and demented older adults. Journal of Cognitive Neuroscience, 22(12), 2677-2684.

[2] Young, V. M., et al. (2025). Data leakage in deep learning for Alzheimer's disease: a scoping review. Diagnostics, 15(18), 2348. https://www.mdpi.com/2075-4418/15/18/2348

[3] Lakshminarayanan, B., Pritzel, A., & Blundell, C. (2017). Simple and scalable predictive uncertainty estimation using deep ensembles. Advances in Neural Information Processing Systems 30 (NeurIPS).

[4] Conformal prediction in clinical artificial intelligence: enhancing model reliability and interpretability. CHEST, 2025. https://journal.chestnet.org/article/S0012-3692(25)05184-0/fulltext

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